HARMONIC COMPANION ROWS FOR BINOMIAL AND APÉRY-LIKE SUMS: CLOSED FORMS, LUCAS DESCENT, AND THE $\zeta(2)$ BOUNDARY

Abstract. For $d \geq 1$ let $A_d(n) = \sum_k \binom{n}{k}^d$. We prove a Lucas theorem for every even coefficient of the Straub–Zudilin deformation of A_d. In particular, its normalized quadratic coefficient is

$$B_d(n) = \sum_{k=0}^{n} \binom{n}{k}^d \left\{ \frac{H_k^{(2)} + H_{n-k}^{(2)}}{2(d+1)} + \frac{d(H_k - H_{n-k})^2}{2(d+1)} \right\}.$$

and satisfies the uniform digit law

$$p^2 B_d(ap+r) \equiv B_d(a) A_d(r) \pmod{p} \qquad (0 \leq a, r < p,\ p \nmid 2(d+1)).$$

More generally the coefficient of t^{2j} satisfies a p^{2j} digit law, with an explicit weight given by a complete Bell polynomial in harmonic numbers. For $d = 3$ and 4, finite certificates identify the quadratic rows with the canonical second solutions of the Franel and s_{10} recurrences. Both identifications and their recurrence-level Lucas laws have also been checked in Lean, including every boundary cell. We explain the rank-two harmonic certificate mechanism and derive exact Casoratian series, alternating error bounds, and a polynomial continued fraction for $5/\zeta(2)$. We compare the result with the minimal weight-three formula for Apéry's $\zeta(3)$ companion, and settle the previously open minimal $\zeta(2)$ formula. We also obtain a new five-term harmonic companion for Cooper's sporadic sequence s_7. It proves a p^2 companion Lucas law, the Apéry limit $\zeta(2)/7$, and an associated polynomial continued fraction. The minimal Apéry proof lifts the harmonic weight to a proper two-parameter hypergeometric deformation, differentiates an order-three creative telescope, and eliminates the auxiliary derivatives in the Ore algebra of shifts.

1. Introduction and main results

Put

$$H_m^{(q)} = \sum_{j=1}^{m} \frac{1}{j^q}, \qquad H_0^{(q)} = 0,$$

and write $H_m = H_m^{(1)}$. The sums

$$A_d(n) = \sum_{k=0}^{n} \binom{n}{k}^d$$

include the Franel numbers at $d = 3$ and the sporadic sequence usually denoted s_{10} at $d = 4$. Straub and Zudilin [10] embedded these rows in an analytic deformation and determined its Apéry limits. Its normalized quadratic coefficient is the following entirely finite sum.

Definition 1.1 (harmonic companion row)**.** For $d \geq 1$, set

$$B_d(n) = \sum_{k=0}^{n} \binom{n}{k}^d w_d(n,k), \qquad w_d(n,k) = \frac{\sigma(n,k) + d\delta(n,k)^2}{2(d+1)}, \tag{1}$$

where

$$\sigma(n,k) = H_k^{(2)} + H_{n-k}^{(2)}, \qquad \delta(n,k) = H_k - H_{n-k}. \tag{2}$$

Thus $B_d(0) = 0$ and $B_d(1) = 1$.

Our first result does not require a recurrence.

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Theorem 1.2 (uniform Lucas law)**.** *Let $d \geq 1$, let p be a prime with $p \nmid 2(d+1)$, and let $0 \leq a, r < p$. Then $p^2 B_d(ap+r)$ is p-integral and*

$$p^2 B_d(ap+r) \equiv B_d(a) A_d(r) \pmod{p\mathbb{Z}_{(p)}}. \tag{3}$$

Here and below a rational congruence $x \equiv y \pmod{p\mathbb{Z}_{(p)}}$ means that both numbers are in the localization $\mathbb{Z}_{(p)}$ and $x - y \in p\mathbb{Z}_{(p)}$. This convention is essential because harmonic numbers need not themselves be p-integral.

For $d = 3, 4$ the companion rows are genuine second solutions of order-two recurrences.

Theorem 1.3 (Franel companion)**.** *Let $F_n = \sum_k \binom{n}{k}^3$ and let $C_0 = 0, C_1 = 1$ satisfy*

$$(n+1)^2 C_{n+1} = (7n^2 + 7n + 2) C_n + 8n^2 C_{n-1} \qquad (n \geq 1). \tag{4}$$

Then

$$C_n = \sum_{k=0}^{n} \binom{n}{k}^3 \left\{ \frac{H_k^{(2)} + H_{n-k}^{(2)}}{8} + \frac{3}{8}(H_k - H_{n-k})^2 \right\} = B_3(n). \tag{5}$$

For every odd prime p and $0 \leq a, r < p$,

$$p^2 C_{ap+r} \equiv C_a F_r \pmod{p\mathbb{Z}_{(p)}}. \tag{6}$$

Theorem 1.4 (s_{10} companion)**.** *Let $S_n = \sum_k \binom{n}{k}^4$ and let $D_0 = 0, D_1 = 1$ satisfy*

$$(n+1)^3 D_{n+1} = (2n+1)(6n^2 + 6n + 2) D_n + n(64n^2 - 4) D_{n-1}. \tag{7}$$

Then

$$D_n = \sum_{k=0}^{n} \binom{n}{k}^4 \left\{ \frac{H_k^{(2)} + H_{n-k}^{(2)}}{10} + \frac{2}{5}(H_k - H_{n-k})^2 \right\} = B_4(n). \tag{8}$$

If $p \nmid 10$ and $0 \leq a, r < p$, then

$$p^2 D_{ap+r} \equiv D_a S_r \pmod{p\mathbb{Z}_{(p)}}. \tag{9}$$

The harmonic formulas in Theorems 1.3 and 1.4 are the $d = 3, 4$ specializations of the known deformation coefficient. The additions here are (i) the direct Lucas descent for the normalized companion row, (ii) explicit finite all-boundary recurrence certificates in these two cases, and (iii) a machine-checked identification with the recurrence-defined second solutions.

2. The deformation and the universal weight

Consider the Straub–Zudilin deformation

$$\mathcal{A}_d(n,t) = \sum_{k=0}^{n} \binom{n}{k}^d \prod_{j=1}^{k} \left(1 - \frac{t}{j}\right)^{-d} \prod_{j=1}^{n-k} \left(1 + \frac{t}{j}\right)^{-d}. \tag{10}$$

The substitution $k \mapsto n-k$, $t \mapsto -t$ shows that the sum is even in t. For a fixed summand, logarithmic expansion gives

$$\begin{aligned} &\prod_{j=1}^{k}(1 - t/j)^{-d} \prod_{j=1}^{n-k}(1 + t/j)^{-d} \\ &\quad = \exp\left(dt\delta + \frac{d}{2}t^2\sigma + O(t^3)\right) = 1 + dt\delta + \frac{t^2}{2}\left(d\sigma + d^2\delta^2\right) + O(t^3). \end{aligned}$$

Consequently, if $\mathcal{A}_{d,1}(n)$ denotes the coefficient of t^2 in (10), then

$$\mathcal{A}_{d,1}(n) = d(d+1) B_d(n). \tag{11}$$

The normalization is forced by $\mathcal{A}_{d,1}(1) = d(d+1)$.

Straub and Zudilin proved that deformation coefficients solve the telescoping recurrences attached to A_d in their stated stable range and that

$$\lim_{n\to\infty} \frac{B_d(n)}{A_d(n)} = \frac{\zeta(2)}{d+1}. \tag{12}$$

The limit is also transparent directly from (1): the binomial-power measure concentrates at $k = n/2$, where $\sigma(n,k) \to 2\zeta(2)$ and $\delta(n,k) \to 0$. Formula (12) is contextual rather than a new claim here.

For $d \geq 5$ the minimal recurrence for A_d need not have order two, so the phrase "second solution" is no longer canonical. Definition 1.1 and Theorem 1.2, however, remain meaningful for every d.

3. Proof of the universal Lucas law

The proof uses only ordinary Lucas congruence and harmonic digit descent.

Lemma 3.1 (harmonic descent)**.** *For a prime p, $q \geq 1$, $b \geq 0$, and $0 \leq s < p$,*

$$p^q H^{(q)}_{bp+s} \in \mathbb{Z}_{(p)}, \qquad p^q H^{(q)}_{bp+s} \equiv H^{(q)}_b \pmod{p\mathbb{Z}_{(p)}}. \tag{13}$$

Proof. Separate the summation indices divisible by p:

$$p^q \sum_{j=1}^{bp+s} \frac{1}{j^q} = \sum_{i=1}^{b} \frac{1}{i^q} + \sum_{\substack{1\leq j\leq bp+s \\ p\nmid j}} \frac{p^q}{j^q}.$$

Every term in the second sum belongs to $p\mathbb{Z}_{(p)}$. □

Proof of Theorem 1.2. Write every $k \leq ap+r$ uniquely as $k = bp+s$, with $0 \leq s < p$. By Lemma 3.1, $p^2 w_d(ap+r,k)$ is p-integral. If $s > r$, Lucas' theorem gives $p \mid \binom{ap+r}{bp+s}$; hence

$$\binom{ap+r}{bp+s}^d p^2 w_d(ap+r, bp+s) \equiv 0 \pmod{p\mathbb{Z}_{(p)}}.$$

If $s \leq r$, subtraction has no borrow:

$$ap + r - (bp + s) = (a-b)p + (r-s).$$

Lucas' theorem and Lemma 3.1 now give

$$\binom{ap+r}{bp+s}^d \equiv \binom{a}{b}^d \binom{r}{s}^d \pmod{p},$$

$$p^2\sigma(ap+r, bp+s) \equiv H^{(2)}_b + H^{(2)}_{a-b} \pmod{p\mathbb{Z}_{(p)}},$$

$$p^2\delta(ap+r, bp+s)^2 \equiv (H_b - H_{a-b})^2 \pmod{p\mathbb{Z}_{(p)}}.$$

Because $2(d+1)$ is a unit in $\mathbb{Z}_{(p)}$, the last two lines say $p^2 w_d(ap+r, bp+s) \equiv w_d(a,b)$. Summing the surviving digit rectangle $0 \leq b \leq a$, $0 \leq s \leq r$ factors the result:

$$p^2 B_d(ap+r) \equiv \sum_{b=0}^{a}\sum_{s=0}^{r} \binom{a}{b}^d \binom{r}{s}^d w_d(a,b) = B_d(a)A_d(r).$$

□

Remark 3.2. The proof explains both the exponent p^2 and the shape of the answer. Each monomial in the weight is homogeneous of harmonic weight two, so multiplication by p^2 extracts its high digit. Cells with a low-digit borrow disappear through the binomial shell; the remaining rectangle separates into high and low digit sums.

4. The full higher-coefficient Lucas tower

The quadratic row is the first case of a theorem for every coefficient of the deformation. Put

$$\mathcal{L}_q(n,k) = H_k^{(q)} + (-1)^q H_{n-k}^{(q)} \tag{14}$$

and define the homogeneous polynomial

$$\mathcal{Q}_{d,m}(n,k) = [t^m] \exp\left(d \sum_{q\geq 1} \frac{\mathcal{L}_q(n,k)}{q}, t^q \right). \tag{15}$$

Equivalently,

$$\mathcal{Q}_{d,m} = \sum_{m_1+2m_2+\cdots+m m_m = m} \prod_{q=1}^{m} \frac{1}{m_q!} \left(\frac{d\mathcal{L}_q}{q} \right)^{m_q}. \tag{16}$$

Every monomial on the right has total harmonic weight m. By logarithmic expansion of (10), its coefficient rows are

$$A_{d,j}^{\langle 2\rangle}(n) := [t^{2j}]\mathcal{A}_d(n,t) = \sum_{k=0}^{n} \binom{n}{k}^d \mathcal{Q}_{d,2j}(n,k). \tag{17}$$

Theorem 4.1 (higher-coefficient Lucas tower)**.** *Let $d, j \geq 1$, let $p > 2j$ be prime, and let $0 \leq a, r < p$. Then*

$$p^{2j} A_{d,j}^{\langle 2\rangle}(ap + r) \equiv A_{d,j}^{\langle 2\rangle}(a) A_d(r) \pmod{p\mathbb{Z}_{(p)}}. \tag{18}$$

If

$$c_{d,j} := A_{d,j}^{\langle 2\rangle}(1) = 2\binom{d+2j-1}{2j} \tag{19}$$

is a p-adic unit, the normalized row $\widehat{A}_{d,j} = A_{d,j}^{\langle 2\rangle}/c_{d,j}$ satisfies

$$p^{2j} \widehat{A}_{d,j}(ap + r) \equiv \widehat{A}_{d,j}(a) A_d(r) \pmod{p\mathbb{Z}_{(p)}}. \tag{20}$$

Proof. For $q \leq 2j$, Lemma 3.1 gives

$$p^q \mathcal{L}_q(ap + r, bp + s) \equiv \mathcal{L}_q(a, b) \pmod{p\mathbb{Z}_{(p)}}$$

whenever $s \leq r$, since then $ap + r - (bp + s) = (a - b)p + (r - s)$. Formula (16) is homogeneous of weight $2j$; its denominators q and $m_q!$ are p-adic units because $p > 2j$. Hence

$$p^{2j} \mathcal{Q}_{d,2j}(ap + r, bp + s) \equiv \mathcal{Q}_{d,2j}(a, b) \pmod{p\mathbb{Z}_{(p)}}. \tag{21}$$

For a borrow cell $s > r$, $p^{2j}\mathcal{Q}_{d,2j}$ is p-integral and Lucas' theorem makes $\binom{ap+r}{bp+s}^d$ divisible by p. Those cells therefore vanish. On the remaining rectangle Lucas' theorem and (21) give

$$\begin{aligned} p^{2j} A_{d,j}^{\langle 2\rangle}(ap + r) &\equiv \sum_{b=0}^{a} \sum_{s=0}^{r} \binom{a}{b}^d \binom{r}{s}^d \mathcal{Q}_{d,2j}(a, b) \\ &= A_{d,j}^{\langle 2\rangle}(a) A_d(r). \end{aligned}$$

At $n = 1$, equation (10) is $(1 - t)^{-d} + (1 + t)^{-d}$, whose t^{2j} coefficient is $2\binom{d+2j-1}{2j}$. Division by a p-adic unit proves the normalized statement. □

The coefficientwise tower has a useful generating-series form. Adopt the convention $A_{d,0}^{\langle 2\rangle} = A_d$ and, for a prime p, put

$$\mathcal{F}_{d,p}^{[J]}(n; T) = \sum_{j=0}^{J} p^{2j} A_{d,j}^{\langle 2\rangle}(n) T^{2j}. \tag{22}$$

Corollary 4.2 (truncated generating-series Lucas law)**.** *Let $d, J \geq 1$, let $p > 2J$ be prime, and let $0 \leq a, r < p$. Then, coefficientwise in $(\mathbb{Z}_{(p)}/p\mathbb{Z}_{(p)})[T]$,*

$$\boxed{\mathcal{F}^{[J]}_{d,p}(ap+r;T) \equiv A_d(r)\mathcal{F}^{[J]}_{d,p}(a;T) \pmod{p}.} \tag{23}$$

Thus the rescaled truncated deformation, rather than any single harmonic row, is itself a Lucas object.

Proof. The constant coefficient is the ordinary Lucas congruence for A_d. For $1 \leq j \leq J$, the coefficient of T^{2j} is exactly Theorem 4.1; the common hypothesis $p > 2J$ implies $p > 2j$. □

Theorem 1.2 is the case $j = 1$, because

$$\mathcal{Q}_{d,2} = \frac{d}{2}\sigma + \frac{d^2}{2}\delta^2 = d(d+1)w_d.$$

The first genuinely new weight occurs at $j = 2$. If $\delta_q = H^{(q)}_k - H^{(q)}_{n-k}$ and $\sigma_q = H^{(q)}_k + H^{(q)}_{n-k}$, then

$$\mathcal{Q}_{d,4} = \frac{d}{4}\sigma_4 + \frac{d^2}{3}\delta_1\delta_3 + \frac{d^2}{8}\sigma_2^2 + \frac{d^3}{4}\delta_1^2\sigma_2 + \frac{d^4}{24}\delta_1^4. \tag{24}$$

Corollary 4.3 (the p^4 row)**.** *Let*

$$\widehat{A}_{d,2}(n) = \frac{12}{d(d+1)(d+2)(d+3)}\sum_{k=0}^{n}\binom{n}{k}^d \mathcal{Q}_{d,4}(n,k).$$

For $p > 4$, $0 \leq a, r < p$, and $p \nmid d(d+1)(d+2)(d+3)$,

$$p^4\widehat{A}_{d,2}(ap+r) \equiv \widehat{A}_{d,2}(a)A_d(r) \pmod{p\mathbb{Z}_{(p)}}.$$

Moreover, after Straub–Zudilin [10],

$$\lim_{n\to\infty}\frac{\widehat{A}_{d,2}(n)}{A_d(n)} = \frac{3(5d+2)}{(d+1)(d+2)(d+3)}\zeta(4). \tag{25}$$

For $d \geq 5$ the row solves every telescoping recurrence in the stable range specified in their theorem.

Proof. The congruence is Theorem 4.1. The coefficient of t^4 in $(t/\sin t)^d$ is $d(5d+2)/360$; divide the resulting limit $d(5d+2)\pi^4/360$ by $c_{d,2} = d(d+1)(d+2)(d+3)/12$ and use $\zeta(4) = \pi^4/90$. □

5. Finite recurrence certificates for $d = 3, 4$

The analytic deformation proves recurrence membership in a stable range. For the Franel and s_{10} rows we instead use a finite identity valid at every $n \geq 1$, including the top cells. We summarize the method because it also explains why the harmonic weight in (1) is tractable.

Let $S_d(n,k) = \binom{n}{k}^d$ and let

$$L[u]_n = c_+(n)u_{n+1} + c_0(n)u_n + c_-(n)u_{n-1}.$$

A Zeilberger certificate G is chosen so that

$$L[S_d(\cdot,k)]_n = G(n,k+1) - G(n,k), \qquad G(n,k) = S_d(n,k)\frac{k^d\rho(n,k)}{(n+1-k)^d}. \tag{26}$$

The symmetric letters in (2) obey

$$\sigma(n+1,k) - \sigma(n,k) = (n+1-k)^{-2}, \qquad \delta(n+1,k) - \delta(n,k) = -(n+1-k)^{-1}, \tag{27}$$

$$\sigma(n,k+1) - \sigma(n,k) = (k+1)^{-2} - (n-k)^{-2}, \quad \delta(n,k+1) - \delta(n,k) = (k+1)^{-1} + (n-k)^{-1}. \tag{28}$$

Thus, after Abel summation, every residual term belongs to the rank-two module

$$S_d(n,k)\big(\mathbb{Q}(n,k)+\mathbb{Q}(n,k)\delta(n,k)\big).$$

It is enough to solve two rational Gosper equations and form

$$\Psi(n,k)=S_d(n,k)\big(z_0(n,k)+z_1(n,k)\delta(n,k)\big). \tag{29}$$

The remaining assertion is a finite list of rational identities plus boundary values at $k=0,n,n+1$. Clearing denominators turns each one into a polynomial identity over $\mathbb{Q}$.

For reference, the Franel Zeilberger numerator is

$$\rho_F(n,k)=-\frac{1}{n}\big(4-12k+12k^2-4k^3+22n-39kn+18k^2n \tag{30}$$

$$+32n^2-27kn^2+14n^3\big). \tag{31}$$

The two Gosper numerators may be taken as

$$\begin{aligned}N_F&=\frac{3k^2}{4n}\big(4-16k+24k^2-16k^3+4k^4+26n-68kn+63k^2n-20k^3n\\&\quad+54n^2-88kn^2+39k^2n^2+46n^3-36kn^3+14n^4\big),\\ M_F&=-\frac{k}{2n}\big(2-10k+20k^2-20k^3+10k^4-2k^5+15n-50kn+70k^2n\\&\quad-45k^3n+11k^4n+40n^2-90kn^2+80k^2n^2-25k^3n^2\\&\quad+50n^3-70kn^3+30k^2n^3+30n^4-20kn^4+7n^5\big),\end{aligned}$$

with $z_1=N_F/(n+1-k)^4$ and $z_0=M_F/(n+1-k)^5$. They satisfy the forced top values

$$\rho_F(n,n+1)=-(n+1)^2,\quad N_F(n)=\frac{3}{4}\big((n+1)^5-(n+1)\big),\quad M_F(n)=-\frac{1}{2}\big((n+1)^5+1\big).$$

The $d=4$ certificate has the same architecture but substantially larger polynomials; it is recorded exactly in the accompanying Lean source. Its top values are

$$\rho_{10}(n,n+1)=-(n+1)^3,\quad N_{10}(n)=\frac{4}{5}\big((n+1)^7-(n+1)^2\big),\quad M_{10}(n)=-\frac{1}{2}\big((n+1)^7+(n+1)\big).$$

Equations (26)–(29), the boundary identities, and the initial values prove Theorems 1.3 and 1.4; Theorem 1.2 then gives their Lucas congruences.

The first values provide a useful normalization check:

n	0	1	2	3	4	5
C_n	0	1	4	208/9	1280/9	208384/225

and

n	0	1	2	3	4
D_n	0	1	21/4	1001/18	85085/144.

6. A new harmonic companion for Cooper's s_7

The harmonic-lifting method is not confined to powers of a single binomial coefficient. Consider Cooper's sporadic sequence s_7 [4, 5],

$$A_n^{(7)}=\sum_{k=0}^{n}S_7(n,k),\qquad S_7(n,k)=\binom{n}{k}^2\binom{n+k}{k}\binom{2k}{n}. \tag{32}$$

It satisfies

$$(n+1)^3u_{n+1}=(2n+1)(13n^2+13n+4)u_n+3n(9n^2-1)u_{n-1}. \tag{33}$$

Let $B_0^{(7)}=0, B_1^{(7)}=1$ denote its normalized companion solution.

Theorem 6.1 (the s_7 harmonic companion). *For every $n \geq 0$,*

$$\boxed{B_n^{(7)} = \frac{1}{28} \sum_{k=0}^{n} S_7(n,k) \left\{ 3(H_k - H_{n-k})(2H_{2k} - 3H_k + H_{n-k}) + 5H_k^{(2)} + 2H_n^{(2)} - 3H_{n-k}^{(2)} \right\}.} \tag{34}$$

The proof also gives the auxiliary identity

$$\sum_{k=0}^{n} S_7(n,k)(2H_{2k} - 3H_k + H_{n-k}) = 0. \tag{35}$$

Proof. Put

$$X_m(t) = \frac{m!}{(1-t)_m} = \prod_{j=1}^{m} (1 - t/j)^{-1}$$

and lift the summand to the proper hypergeometric deformation

$$\Phi_n(u,v) = \sum_k S_7(n,k) \frac{X_k(3u + 5v/3)}{X_k(14v/3)} X_{n-k}(-3u+v) X_{2k}(2v) \times \frac{X_n(2u+2v)}{X_n(u+v)^2}. \tag{36}$$

At the origin the two logarithmic derivatives of its summand are

$$\mathcal{A} = 3(H_k - H_{n-k}), \qquad \mathcal{B} = 2H_{2k} - 3H_k + H_{n-k},$$

and its mixed logarithmic derivative is

$$\mathcal{C} = 5H_k^{(2)} + 2H_n^{(2)} - 3H_{n-k}^{(2)}.$$

Consequently the mixed derivative of the summand is $S_7(\mathcal{AB} + \mathcal{C})$, exactly twenty-eight times the summand in (34).

It remains to prove recurrence membership; a single direct harmonic Gosper telescope does not exist. We instead polarize the mixed derivative. Set

$$\Phi_n^u(t) = \Phi_n(t,0), \qquad \Phi_n^v(t) = \Phi_n(0,t), \qquad \Phi_n^d(t) = \Phi_n(t,t).$$

The diagonal is especially small:

$$\Phi_n^d(t) = \sum_k S_7(n,k) \frac{X_{n-k}(-2t) X_{2k}(2t) X_n(4t)}{X_n(2t)^2}. \tag{37}$$

Exact creative telescoping gives parameter-dependent operators of orders $2, 4, 3$ for Φ^u, Φ^v, Φ^d, respectively. Write $C_r^x = \partial_t^r C^x(t)|_{t=0}$ for their operator coefficients and put

$$U = \partial_u \Phi|_0, \quad V = \partial_v \Phi|_0, \quad U_2 = \partial_u^2 \Phi|_0, \quad V_2 = \partial_v^2 \Phi|_0, \quad D_2 = \frac{d^2}{dt^2} \Phi(t,t)\Big|_{t=0}.$$

For the forward operator

$$\mathcal{L}_7 = -3(n+1)(3n+2)(3n+4) - (2n+3)(13n^2 + 39n + 30)E + (n+2)^3 E^2, \tag{38}$$

exact right division gives

$$C_0^u = -7(n+1)^2(n+2)^3 \mathcal{L}_7,$$
$$C_0^v = Q_0 \mathcal{L}_7, \qquad C_1^v = Q_1 \mathcal{L}_7.$$

for explicit order-two rational operators Q_0, Q_1. The first derivative of the v-telescope therefore gives $C_0^v V = 0$. The leading coefficient does not vanish for $n \geq 0$, while $V_0 = V_1 = V_2 = V_3 = 0$; this proves (35).

The second derivatives of the three telescopes now read

$$\begin{aligned} C_0^u U_2 + 2C_1^u U + C_2^u A^{(7)} &= 0,\\ C_0^v V_2 + C_2^v A^{(7)} &= 0,\\ C_0^d D_2 + 2C_1^d U + C_2^d A^{(7)} &= 0. \end{aligned}$$

Eliminating $A^{(7)}$ and U in the Ore algebra gives

$$\mathcal{P}\mathcal{L}_7 J = 0, qquad J = \frac{D_2 - U_2 - V_2}{2} = \partial_u \partial_v \Phi|_0, \tag{39}$$

where $\mathcal{P} = P_0 + P_1 E + P_2 E^2$ and

$$\begin{aligned} P_0 &= 2(n+1)(n+2)(98n^3 + 1029n^2 + 3526n + 3931),\\ P_1 &= (n+2)(392n^4 + 4606n^3 + 19543n^2 + 35229n + 22598),\\ P_2 &= (n+3)(2n+7)(98n^3 + 735n^2 + 1762n + 1336). \end{aligned}$$

All three cell identities, their Taylor coefficients, and the right remainder in (39) are exact rational identities. At integer n the certificate flux has finite support; equivalently, one may regularize n to $n + \varepsilon$ and take the removable limits at the exceptional cells. Thus summing the cell identities introduces no endpoint term.

The order-four operator $\mathcal{P}\mathcal{L}_7$ has nonzero leading coefficient for every $n \geq 0$. Finally,

$$(J_0, J_1, J_2, J_3) = (0, 28, 315, 45010/9) = 28(B_0^{(7)}, B_1^{(7)}, B_2^{(7)}, B_3^{(7)}).$$

Since $\mathcal{L}_7 B^{(7)} = 0$, four initial values prove $J = 28B^{(7)}$. □

Theorem 6.2 (s_7 companion Lucas law)**.** *Let $p \nmid 28$ be prime and let $0 \leq a, r < p$. Then*

$$p^2 B_{ap+r}^{(7)} \equiv B_a^{(7)} A_r^{(7)} \pmod{p\mathbb{Z}_{(p)}}. \tag{40}$$

Proof. Write $k = bp + s$, $0 \leq s < p$. The three binomial factors in (32) give the termwise Lucas congruence

$$S_7(ap + r, bp + s) \equiv S_7(a, b) S_7(r, s) \pmod{p}. \tag{41}$$

Indeed, if the left side is nonzero modulo p, the low digit of $\binom{ap+r}{bp+s}$ forces $s \leq r$, while that of $\binom{(a+b)p+r+s}{bp+s}$ forces $r + s < p$. Hence $2s < p$ as well, and ordinary Lucas factorization applies without a low-digit carry to all three factors. If either condition fails, the corresponding low-digit factor and the right side of (41) both vanish.

Let $w_7(n, k)$ denote the expression in braces in (34), divided by 28. Lemma 3.1 shows that $p^2 w_7(ap + r, bp + s)$ is p-integral. On a nonvanishing Lucas cell we have $s \leq r$ and $2s < p$, so the same lemma, now for the indices $k, n - k, 2k, n$, gives

$$p^2 w_7(ap + r, bp + s) \equiv w_7(a, b) \pmod{p\mathbb{Z}_{(p)}}.$$

All other cells vanish modulo p. Summing the surviving digit rectangle and using (41) gives

$$p^2 B_{ap+r}^{(7)} \equiv \sum_{b=0}^{a} \sum_{s=0}^{r} S_7(a, b) S_7(r, s) w_7(a, b) = B_a^{(7)} A_r^{(7)}.$$

□

Corollary 6.3 (the s_7 Apéry limit)**.**

$$\lim_{n \to \infty} \frac{B_n^{(7)}}{A_n^{(7)}} = \frac{\zeta(2)}{7}. \tag{42}$$

Proof. The factorial form of the shell is

$$S_7(n,k) = \frac{(n+k)!(2k)!}{k!^3(n-k)!^2(2k-n)!}.$$

Its consecutive quotient is

$$\frac{S_7(n,k+1)}{S_7(n,k)} = \frac{(n+k+1)(2k+2)(2k+1)(n-k)^2}{(k+1)^3(2k-n+1)(2k-n+2)}.$$

For $k/n \to x \in (1/2,1)$ this tends to $4(1+x)(1-x)^2/[x(2x-1)^2]$, and the equation that this limit equal one reduces to $4-5x=0$. The quotient bounds therefore give exponential concentration at $k/n = 4/5$.

At that saddle,

$$H_k - H_{n-k} \longrightarrow \log 4, qquad 2H_{2k} - 3H_k + H_{n-k} \longrightarrow 0, qquad H_m^{(2)} \longrightarrow \zeta(2)$$

for each linearly growing index m. The complement of a fixed saddle neighborhood is exponentially negligible, even after the $O(\log^2 n)$ harmonic weight is included. Substitution in (34) leaves $(5+2-3)\zeta(2)/28 = \zeta(2)/7$. □

7. Formal verification

The Franel and s_{10} recurrence proofs and their two specialized Lucas laws have been formalized in Lean 4 with Mathlib. The s_7 theorem is presently verified by the exact symbolic audit described above, but not yet in Lean. The relevant declarations in `lean/ZetaLucas/FranelClosedForm.lean` are

`franel_harmonic_closed_form`,	`s10_harmonic_closed_form`,
`franelB_lucas`,	`s10B_lucas`.

The proof does not infer recurrence membership from a finite sample. Lean checks the generic cleared-denominator identities, the exceptional boundary cells, the telescoping finite sums, and the initial-value identification. The Lucas statements are transferred to the recurrence-defined sequences only after the harmonic sums have been identified with those sequences.

The target reproduces with

```
cd lean
lake build ZetaLucas.FranelClosedForm
```

and contains no `sorry`, `admit`, custom axiom, or unsafe declaration. The axiom audit of the four headline theorems reports only Mathlib's standard logical axioms `propext`, `Classical.choice`, and `Quot.sound`.

8. Casoratians, certified brackets, and continued fractions

The closed forms identify the companion rows, but the recurrence alone then supplies another exact description. We record the general observation first.

Lemma 8.1 (Casoratian and alternating error)**.** *Suppose positive sequences A_n and B_n satisfy*

$$\alpha_n u_{n+1} = \beta_n u_n + \gamma_n u_{n-1} \qquad (n \geq 1), \tag{43}$$

where $\alpha_n, \beta_n, \gamma_n > 0$, and suppose $A_0 = 1$, $B_0 = 0$, $B_1 = 1$. Put

$$W_n = A_n B_{n+1} - A_{n+1} B_n.$$

Then

$$W_0 = 1, \qquad W_n = -\frac{\gamma_n}{\alpha_n} W_{n-1}, \tag{44}$$

and

$$\frac{B_{n+1}}{A_{n+1}} - \frac{B_n}{A_n} = \frac{W_n}{A_n A_{n+1}}. \tag{45}$$

If $B_n/A_n \to L$*, then*

$$L = \sum_{m=0}^{\infty} \frac{W_m}{A_m A_{m+1}}. \tag{46}$$

Moreover, with $R_n = B_n/A_n$ *and* $\varepsilon_n = |W_n|/(A_n A_{n+1})$*,*

$$\begin{cases} R_n < L < R_n + \varepsilon_n, & n \text{ even}, \\ R_n - \varepsilon_n < L < R_n, & n \text{ odd}. \end{cases} \tag{47}$$

Proof. Substitute the recurrence for A_{n+1} and B_{n+1} into the determinant; the β_n terms cancel and give (44). Division by $A_n A_{n+1}$ gives (45), which telescopes. The terms alternate by (44), and their absolute values strictly decrease because

$$\frac{\varepsilon_n}{\varepsilon_{n-1}} = \frac{\gamma_n}{\alpha_n} \frac{A_{n-1}}{A_{n+1}} < 1;$$

the strict inequality is the positive $\beta_n A_n$ term in (43). They tend to zero since the partial sums converge to L. The alternating-series remainder theorem proves (47). □

For the Franel recurrence, Lemma 8.1 gives

$$F_n C_{n+1} - F_{n+1} C_n = \frac{(-8)^n}{(n+1)^2}. \tag{48}$$

Consequently,

$$\boxed{\zeta(2) = 4 \sum_{n=0}^{\infty} \frac{(-8)^n}{(n+1)^2 F_n F_{n+1}}.} \tag{49}$$

This identity, and its connection with a continued fraction discovered by the Ramanujan Machine, is proved in [2]. Here it also follows at once from Theorem 1.3 and its Apéry limit. The additional conclusion (47) gives certified rational intervals: if

$$R_n^F = \frac{C_n}{F_n}, \qquad \varepsilon_n^F = \frac{8^n}{(n+1)^2 F_n F_{n+1}},$$

then for even n, $4R_n^F < \zeta(2) < 4(R_n^F + \varepsilon_n^F)$, with the odd case given by the second line of (47).

The same calculation for s_{10} appears not to have been recorded in the sources located for this paper.

Theorem 8.2 (s_{10} Casoratian and Basel series)**.** *With* S_n, D_n *as in Theorem 1.4,*

$$S_n D_{n+1} - S_{n+1} D_n = (-64)^n \frac{(1)_n (3/4)_n (5/4)_n}{(2)_n^3}. \tag{50}$$

Hence

$$\boxed{\zeta(2) = 5 \sum_{n=0}^{\infty} \frac{(-64)^n (1)_n (3/4)_n (5/4)_n}{(2)_n^3 S_n S_{n+1}}.} \tag{51}$$

The partial ratios D_n/S_n *give the strict alternating brackets* (47) *for* $\zeta(2)/5$.

Proof. Here

$$\frac{W_n}{W_{n-1}} = -\frac{n(64n^2-4)}{(n+1)^3} = -64\frac{n(n-1/4)(n+1/4)}{(n+1)^3}.$$

Iterating from $W_0 = 1$ proves (50). Equations (51) and the brackets follow from Lemma 8.1 and $D_n/S_n \to \zeta(2)/5$. □

Theorem 8.3 (s_{10} polynomial continued fraction)**.** *Set*

$$a_n = (2n+1)(6n^2+6n+2), \qquad b_n = n^4(64n^2-4).$$

Then

$$\boxed{\frac{5}{\zeta(2)} = a_0 + \cfrac{b_1}{a_1 + \cfrac{b_2}{a_2 + \cfrac{b_3}{a_3 + \ddots}}}.} \tag{52}$$

The nth convergent is exactly S_{n+1}/D_{n+1}.

Proof. Let $P_{-2} = 0, P_{-1} = 1$ and $Q_{-1} = 0, Q_0 = 1$ be the Euler–Wallis numerator and denominator sequences, so

$$P_n = a_nP_{n-1} + b_nP_{n-2} \quad (n \geq 0), \qquad Q_n = a_nQ_{n-1} + b_nQ_{n-2} \quad (n \geq 1).$$

Multiplying (7) by $n!^3$ shows, including the initial cells,

$$P_n = (n+1)!^3S_{n+1}, \qquad Q_n = (n+1)!^3D_{n+1}.$$

Thus the nth convergent is $P_n/Q_n = S_{n+1}/D_{n+1}$, which tends to $5/\zeta(2)$. □

The new s_7 companion supplies a direct harmonic proof of the continued fraction evaluation discussed in [6].

Theorem 8.4 (s_7 Basel series and continued fraction)**.** *With $A_n^{(7)}, B_n^{(7)}$ as in Theorem 6.1,*

$$A_n^{(7)}B_{n+1}^{(7)} - A_{n+1}^{(7)}B_n^{(7)} = (-27)^n\frac{(1)_n(2/3)_n(4/3)_n}{(2)_n^3}. \tag{53}$$

Consequently

$$\boxed{\zeta(2) = 7\sum_{n=0}^{\infty}\frac{(-27)^n(1)_n(2/3)_n(4/3)_n}{(2)_n^3A_n^{(7)}A_{n+1}^{(7)}}.} \tag{54}$$

The partial ratios $B_n^{(7)}/A_n^{(7)}$ give the strict alternating brackets of Lemma 8.1 for $\zeta(2)/7$. Moreover, if

$$a_n = (2n+1)(13n^2+13n+4), \qquad b_n = 3n^4(9n^2-1),$$

then

$$\boxed{\frac{7}{\zeta(2)} = 4 + \cfrac{b_1}{a_1 + \cfrac{b_2}{a_2 + \cfrac{b_3}{a_3 + \ddots}}}.} \tag{55}$$

Its nth convergent is $A_{n+1}^{(7)}/B_{n+1}^{(7)}$.

Proof. Recurrence (33) gives

$$\frac{W_n}{W_{n-1}} = -27\frac{n(n-1/3)(n+1/3)}{(n+1)^3}.$$

Iteration proves (53); Corollary 6.3 and Lemma 8.1 give the series and brackets. Multiplication of (33) by $n!^3$ gives the Euler–Wallis recurrence with partial denominators a_n and partial numerators b_n. Its convergents are the displayed ratios and tend to $7/\zeta(2)$. □

Remark 8.5 (arithmetic strength). Theorem 6.1 gives a direct harmonic proof of the limit and hence of the continued-fraction evaluation, but it does not by itself give a new Apéry proof of the irrationality of $\zeta(2)$. The characteristic equation at infinity of (33) is

$$\lambda^2 - 26\lambda - 27 = (\lambda - 27)(\lambda + 1).$$

The two formal scales are $27^n n^{-3/2}$ and $(-1)^n n^{-3/2}$. Thus the dominant part cancels in $\zeta(2)A_n^{(7)} - 7B_n^{(7)}$, but the remaining linear form is only on the polynomial scale $O(n^{-3/2})$, rather than exponentially small. Meanwhile the evident denominator clearing supplied by (34) is $28\,\mathrm{lcm}(1,\ldots,2n)^2$. After that clearing the linear forms do not tend to zero, so the standard Apéry integrality argument cannot close. What the theorem does supply is the harmonic numerator formula, the companion Lucas law (40), the exponentially convergent Basel series (54), and the strict rational brackets.

9. Comparison with Apéry's $\zeta(3)$ companion

The same program produces a strikingly smaller formula for the numerator row in Apéry's proof of the irrationality of $\zeta(3)$. Let

$$a_n = \sum_{k=0}^{n} \binom{n}{k}^2 \binom{n+k}{k}^2,$$

and let $b_0 = 0, b_1 = 6$ satisfy Apéry's recurrence

$$(n+2)^3 b_{n+2} = (34n^3 + 153n^2 + 231n + 117)b_{n+1} - (n+1)^3 b_n. \tag{56}$$

Then the minimal harmonic form is

$$b_n = \sum_{k=0}^{n} \binom{n}{k}^2 \binom{n+k}{k}^2 \left(2H_n^{(3)} - H_k^{(3)}\right). \tag{57}$$

An explicit certificate proof and a Lean formalization establish this identity. Its Lucas descent is the weight-three counterpart of Theorem 1.2: multiplication by p^3 extracts the high digit of the harmonic weight.

There is an instructive contrast. The binomial-power companions use the symmetric rank-two algebra generated by 1 and $\delta = H_k - H_{n-k}$ after differencing. Formula (57) is even smaller: it is linear in order-three harmonic numbers and needs no nested tail. Neither simplification survives unchanged in the $\zeta(2)$ Apéry pair.

10. The $\zeta(2)$ companion: two closed forms

Let

$$T(n,k) = \binom{n}{k}^2 \binom{n+k}{k}, \qquad a_n = \sum_{k=0}^{n} T(n,k), \tag{58}$$

and define

$$(Lu)_n = (n+1)^2 u_{n+1} - (11n^2 + 11n + 3)u_n - n^2 u_{n-1}. \tag{59}$$

Thus $La = 0$. Let $B_0 = 0, B_1 = 1$ and $LB = 0$; classically $B_n/a_n \to \zeta(2)/5$.

10.1. **A proved nested-tail closed form.** Put

$$e(n)=\sum_{m=1}^{n}\frac{(-1)^{m-1}}{m^2},\qquad \tau(n,k)=\sum_{m=1}^{\min(k,n)}\frac{(-1)^{n+m-1}}{m^2\binom{n}{m}\binom{n+m}{m}}.$$

Theorem 10.1 ($\zeta(2)$ nested-tail formula)**.** *For every $n\geq 0$,*

$$5B_n=\sum_{k=0}^{n}T(n,k)\big(2e(n)+\tau(n,k)\big). \tag{60}$$

The proof is an explicit certificate chain. Its decisive cancellation can be stated compactly. If

$$\rho(n,k)=k^2+k(1+6n)-4-15n-11n^2,$$

then the Zeilberger and Abel residual simplifies to

$$W(n,k)=3-\frac{3(n+1)}{n+1-k}. \tag{61}$$

The apparent double pole at $k=n+1$ and the pole at $k=-n$ cancel. The last sum is then the elementary binomial identity

$$\sum_{k=0}^{n}\frac{(-1)^k\binom{n}{k}}{n+1-k}=\frac{(-1)^n}{n+1}.$$

The mechanism behind (61) is the tail shift

$$m^2+(n+1-m)(n+1+m)=(n+1)^2,$$

which makes the inner m-sum telescope. This proves recurrence membership at all cells and, with the two initial values, proves Theorem 10.1.

10.2. **The minimal formula.** The substantially shorter depth-one form is

$$B_n=\frac{1}{5}\sum_{k=0}^{n}T(n,k)\left[H_n^{(2)}+H_k\big(2H_k-H_{n-k}-H_n\big)\right]. \tag{62}$$

Define

$$A_n^{\star}=\sum_{k=0}^{n}T(n,k)H_k(2H_k-H_{n-k}-H_n).$$

Theorem 10.2 (minimal $\zeta(2)$ companion)**.** *For every $n\geq 0$, formula (62) holds. Equivalently,*

$$(LA^{\star})_n=-a_{n+1}-a_{n-1}\qquad (n\geq 1). \tag{63}$$

Proof. The harmonic shifts give the exact identity

$$L[H_n^{(2)}a_n]=a_{n+1}+a_{n-1}.$$

Hence applying L to the right side of (62) yields zero precisely when (63) holds. The initial values agree.

We prove (63) by moving the harmonic letters back into parameters. Set

$$X_m(t)=\frac{m!}{(1-t)_m}=\prod_{j=1}^{m}\left(1-\frac{t}{j}\right)^{-1} \tag{64}$$

and introduce the proper hypergeometric sum

$$\Phi_n(u,v)=\sum_{k=0}^{n}T(n,k)X_k(u)\frac{X_k(v)^2}{X_{n-k}(v)X_n(v)}. \tag{65}$$

At the origin its relevant Taylor coefficients are

$$\Phi_n = a_n, \quad \partial_u \Phi_n = U_n, \quad \partial_v \Phi_n = V_n, \quad \partial_u\partial_v \Phi_n = A_n^\star, \tag{66}$$

where every derivative is evaluated at $(u, v) = (0, 0)$ and

$$U_n = \sum_k T(n,k) H_k, \qquad V_n = \sum_k T(n,k)(2H_k - H_{n-k} - H_n).$$

Let E denote the forward shift in n and write

$$\mathcal{L} = -(n+1)^2 - (11n^2 + 33n + 25)E + (n+2)^2 E^2. \tag{67}$$

Thus $(\mathcal{L}u)_n = (Lu)_{n+1}$. Creative telescoping applied directly to the summand of (65) gives an order-three operator

$$\mathcal{C}(u,v) = \sum_{i=0}^{3} C_i(n; u, v) E^i \tag{68}$$

and a rational certificate $R(n, k; u, v)$ satisfying

$$\sum_{i=0}^{3} C_i(n; u, v) F(n+i, k; u, v) = \Delta_k\big(R(n, k; u, v) F(n, k; u, v)\big). \tag{69}$$

The exact certificate is generated and independently rechecked in `work/z2cf/parameter_proof.mac`. Its denominator in k is $(n-k+1)^3(n-k+2)^3(n-k+3)^3$. Summing (69) over the natural finite ranges produces ten top cells and two certificate endpoints. Their coefficients of $1, u, v, uv$ reduce respectively to

$$0, \qquad 0, \qquad 0, \qquad 0. \tag{70}$$

At the origin the operator and its v-derivative factor on the right:

$$\mathcal{C}(0,0) = Q_0\mathcal{L}, qquad \partial_v\mathcal{C}(0,0) = Q_v\mathcal{L}, \tag{71}$$

where

$$Q_0 = -2(2n+3)(29n+69) + (n+3)(29n+40)E, \tag{72}$$

$$Q_v = \frac{204n^3 + 1221n^2 + 2320n + 1383}{(n+1)(n+3)} - \frac{138n^2 + 523n + 468}{n+2} E. \tag{73}$$

The v-derivative of the telescoped relation gives $Q_0\mathcal{L}V = 0$, because $\mathcal{L}a = 0$. Directly $V_0 = V_1 = V_2 = 0$, and the leading coefficient $(n+3)^3(29n+40)$ never vanishes for $n \geq 0$; hence

$$V_n = 0 \qquad (n \geq 0). \tag{74}$$

Write $C_u = \partial_u\mathcal{C}(0,0)$ and $C_{uv} = \partial_u\partial_v\mathcal{C}(0,0)$. There are first-order rational operators X, Y such that

$$XQ_v = YQ_0. \tag{75}$$

Normalize the coefficient of E in X to 1; its constant coefficient is

$$-\frac{2(n+1)(2n+3)(10092n^5 + 133835n^4 + 700700n^3 + 1808755n^2 + 2300030n + 1151344)}{(n+3)(n+4)(10092n^5 + 83375n^4 + 266280n^3 + 408745n^2 + 299740n + 83112)}. \tag{76}$$

Exact Ore division gives, for a rational operator H,

$$YC_u - XC_{uv} - (XQ_0)J = H\mathcal{L}, qquad J = -(1 + E^2). \tag{77}$$

The computed right remainder is identically $(0, 0, 0)$.

Differentiate the telescoped recurrence first in u, and then in both u, v. Using (74), one obtains

$$Q_0\mathcal{L}U + C_u a = 0, qquad Q_0\mathcal{L}A^\star + Q_v\mathcal{L}U + C_{uv} a = 0.$$

Multiply these equations by Y and X, use (75), and subtract. Equation (77) and $\mathcal{L}a = 0$ then yield

$$XQ_0(\mathcal{L}A^\star - Ja) = 0.$$

The leading coefficient of XQ_0 is $(n+4)(29n+69)$, while direct evaluation at $n = 0, 1$ gives $(\mathcal{L}A^\star - Ja)_0 = (\mathcal{L}A^\star - Ja)_1 = 0$. Thus $\mathcal{L}A^\star = Ja$, which is precisely (63) after shifting the index. This proves Theorem 10.2. □

Status 10.3 (current $\zeta(2)$ status). Both the nested-tail identity (60) and the minimal depth-one identity (62) are now proved by exact symbolic certificates. The full latter theorem is not yet formalized in Lean. Its first formal stage, in `lean/ZetaLucas/Z2Shell.lean`, contains no `sorry` and proves the shell absorption identities, the base Apéry recurrence, and the identity $L(H_n^{(2)}a_n) = a_{n+1} + a_{n-1}$. The remaining stage is the certificate and pre-operator elimination. Earlier exclusions of three fixed harmonic certificate modules remain valid: the successful proof escapes them by telescoping the proper hypergeometric deformation before differentiating.

11. Structural summary and open problems

row	closed-form weight	recurrence proof	Lucas/formal status
Franel $d = 3$	$(\sigma + 3\delta^2)/8$	finite rank-two certificate	p^2 Lucas law; closed form and law in Lean
s_{10}, $d = 4$	$(\sigma + 4\delta^2)/10$	finite rank-two certificate	p^2 Lucas law; closed form and law in Lean
Cooper s_7	equation (34)	three polarized parameter telescopes and Ore elimination	p^2 Lucas law; exact symbolic audit; Lean open
Apéry $\zeta(3)$	$2H_n^{(3)} - H_k^{(3)}$	finite minimal certificate	weight-three Lucas descent; Lean proof
Apéry $\zeta(2)$	nested tail and minimal depth-one forms, proved	finite tail certificate; parameter telescope and Ore elimination	exact symbolic audit; Lean formalization open

The immediate questions are:

(1) Formalize Theorems 4.1, 6.1, and 10.2 in Lean, including the parameter telescopes and Ore-operator identities.
(2) Find comparably small all-boundary recurrence certificates for the higher deformation rows when $d \geq 5$.
(3) Strengthen the one-digit congruences to higher powers of p or to a compatible multi-digit theory.
(4) For $d \geq 5$, relate the deformation rows canonically to the higher-order recurrence solution space rather than merely to the deformation coefficient.

Reproducibility record

The formal source is `lean/ZetaLucas/FranelClosedForm.lean`; the human-readable formal audit is `work/FRANEL_S10_LEAN.md`. The full $\zeta(2)$ tail proof and the minimal formula are recorded in `work/z2cf/note.tex`. The exact parameter telescope, all boundary checks, operator factorizations, and Ore remainder are independently recomputed by `work/z2cf/parameter_proof.mac`. For s_7, the three exact telescopes and Taylor certificates are generated by the `work/s7_param/extract_*.mac` scripts; the Ore elimination and right-factor audit are in `work/s7_param/ore_eliminate.py`. Running `work/s7_param/verify.sh` rebuilds all of them and also performs an independent exact check of the first fifty rows. All numerical checks mentioned in this paper are supplementary evidence only; no theorem labeled "proved" uses finite checking as a proof step.

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